

Chapter 6 Part I – Cost Behavior

Shift in Focus

Away
from
GAAP

Product vs. Period Cost

Manufacturing vs. Non-manufacturing Cost

Allocation of indirect costs

Toward
Internal
Decision
Making

Variable versus Fixed Costs (Cost Behavior)

Cost-Volume-Profit

Incremental Analysis

Learning Objectives

Describe key characteristics and graphs of various cost behaviors

Use cost equations to express and predict costs

Understanding cost behavior

Understanding cost behavior is the key to many managerial decisions.

How does total cost change when something else changes, such as:

Producing more units

Serving more customers

Dropping a product line

Outsourcing part of a process

Cost Behavior

Most common cost behaviors:

Variable
costs

Fixed
costs

Mixed
costs

Variable Costs

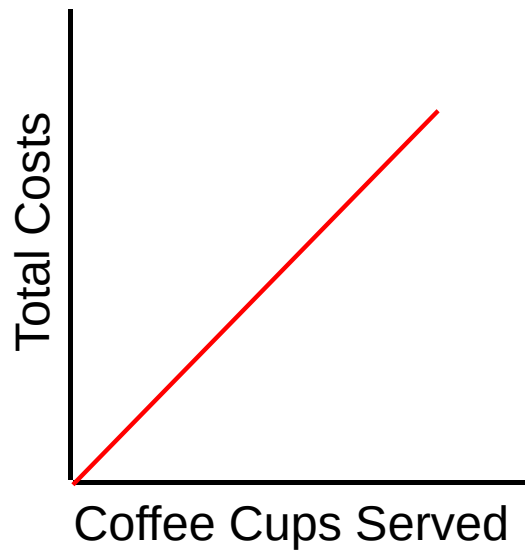
Costs that are incurred for every unit of volume.

Total variable costs change in direct proportion to changes in volume.

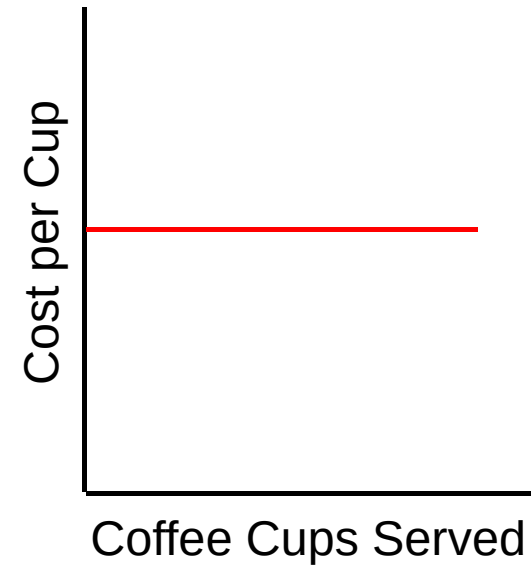
Graphs always begin at the origin.

Slope represents the variable cost per unit of activity.

Variable Costs Graph



Total variable costs increase as activity increases.

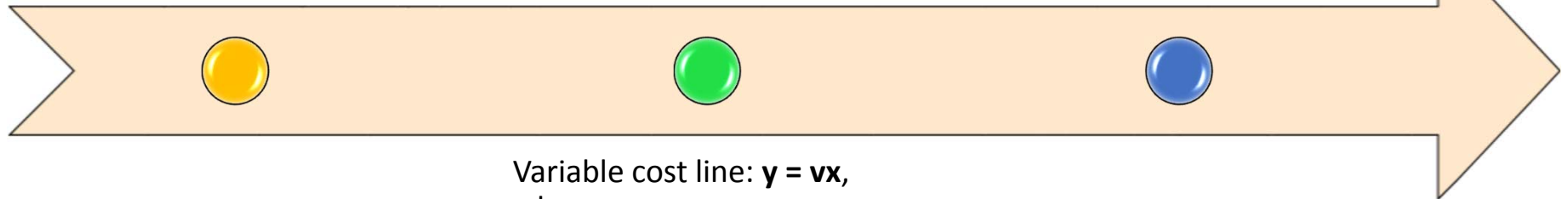


Cost per cup is constant as activity increases.

Cost Equation—Variable Costs

Mathematical equation for a straight line, to express how a cost behaves

Variable cost *per unit* remains constant



Variable cost line: $y = vx$,
where

- y = total variable cost
- v = variable cost per unit of activity
- x = volume of activity

Summarize : Key Characteristics of Variable Costs

- *Total* variable costs change in *direct proportion* to changes in volume
- The *variable cost per unit of activity* (v) remains constant and is the slope of the variable cost line
- Total variable cost graphs always begin at the origin (if volume is zero, total variable costs are zero)
- Total variable costs can be expressed as follows:

$$y = vx$$

where,

y = total variable cost

v = variable cost per unit of activity

x = volume of activity

Fixed Costs

Costs that
do ***not***
change in
total
despite
changes
in volume

Committed fixed
costs

No control over these

Property taxes, insurance, depreciation,
lease payments, etc

Discretionary fixed
costs

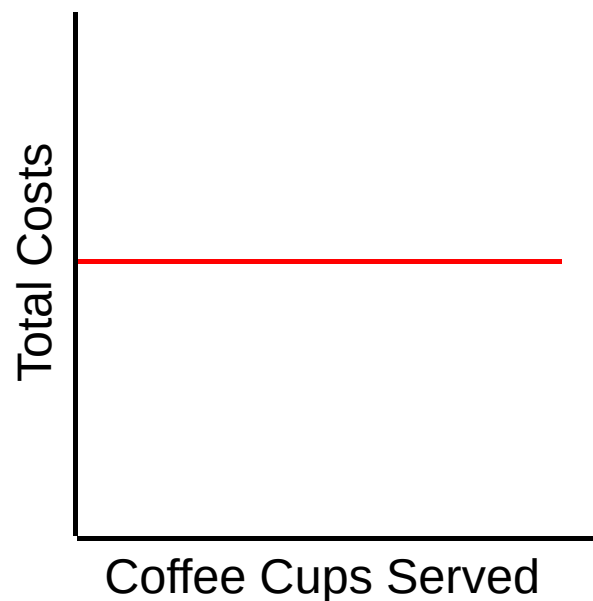
Relatively more control

Manager salaries, advertising expense,
etc.

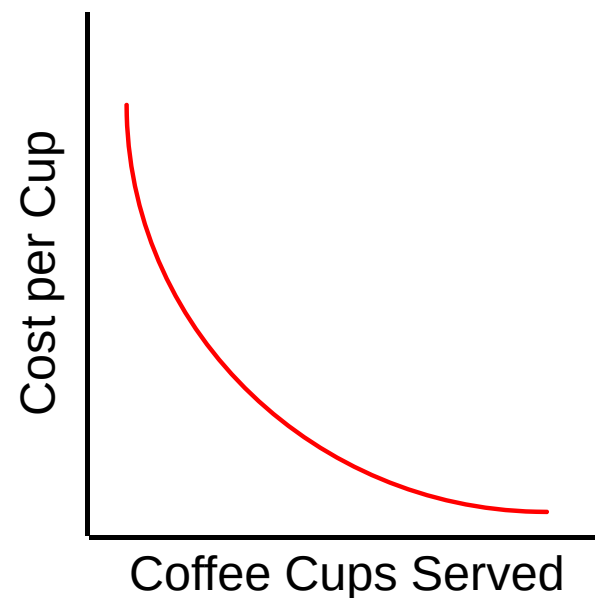
Graph—flat line that
intersects y-axis



Fixed costs



Total fixed costs remain constant as activity increases.



Cost per cup declines as activity increases.

Cost Equation—Fixed Costs

Fixed cost *per unit*
is inversely
proportionate to
volume.

$$y = f, \text{ where}$$

y = total fixed
cost

f = fixed
amount over a
period of time

Summarize - Key Characteristics of Fixed Costs

- Total fixed costs stay *constant* over a wide range of volume
- Fixed costs *per unit of activity* vary *inversely* in proportion to changes in volume:
 - Fixed cost per unit of activity *increases* when volume *decreases*
(If volume is cut in half, the fixed cost per unit will double)
 - Fixed cost per unit of activity *decreases* when volume *increases*
(If volume doubles, the fixed cost per unit will be cut in half)
- Total fixed cost graphs are always flat lines with no slope that intersect the y-axis at a level equal to total fixed costs
- Total fixed costs can be expressed as $y = f$
where,

y = total fixed cost

f = fixed cost over a given period of time

Mixed Costs

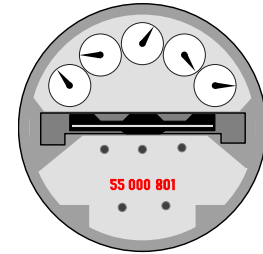
Contain both variable and fixed cost components

Electric Utility bill – cost of wires for transmission → fixed
Electric Utility bill – consumption of electricity → variable

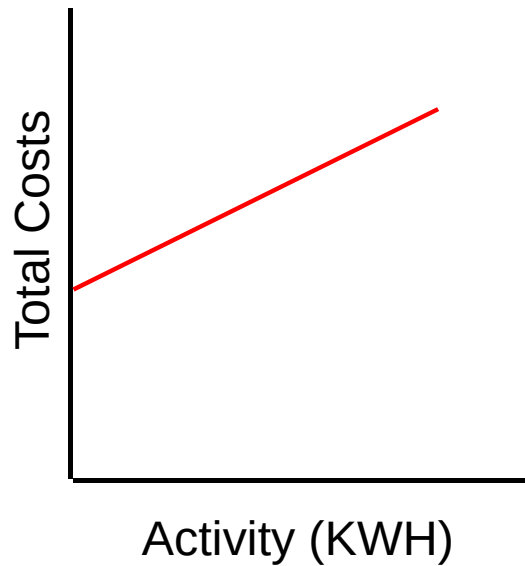
Increases with volume

Does *not* start at the origin

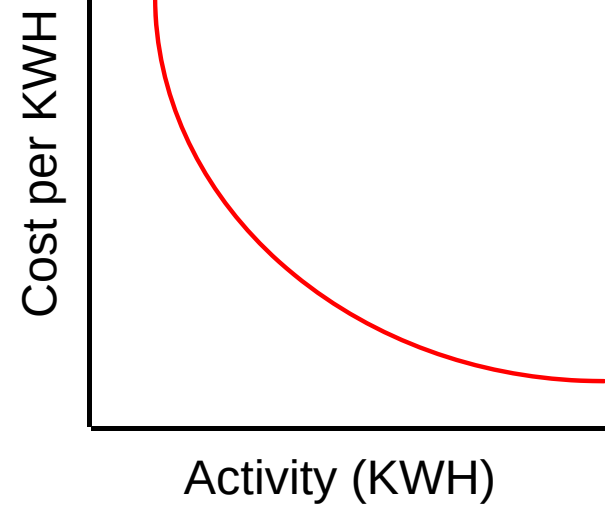
Mixed costs



Example:
Electric
Utility
Bill



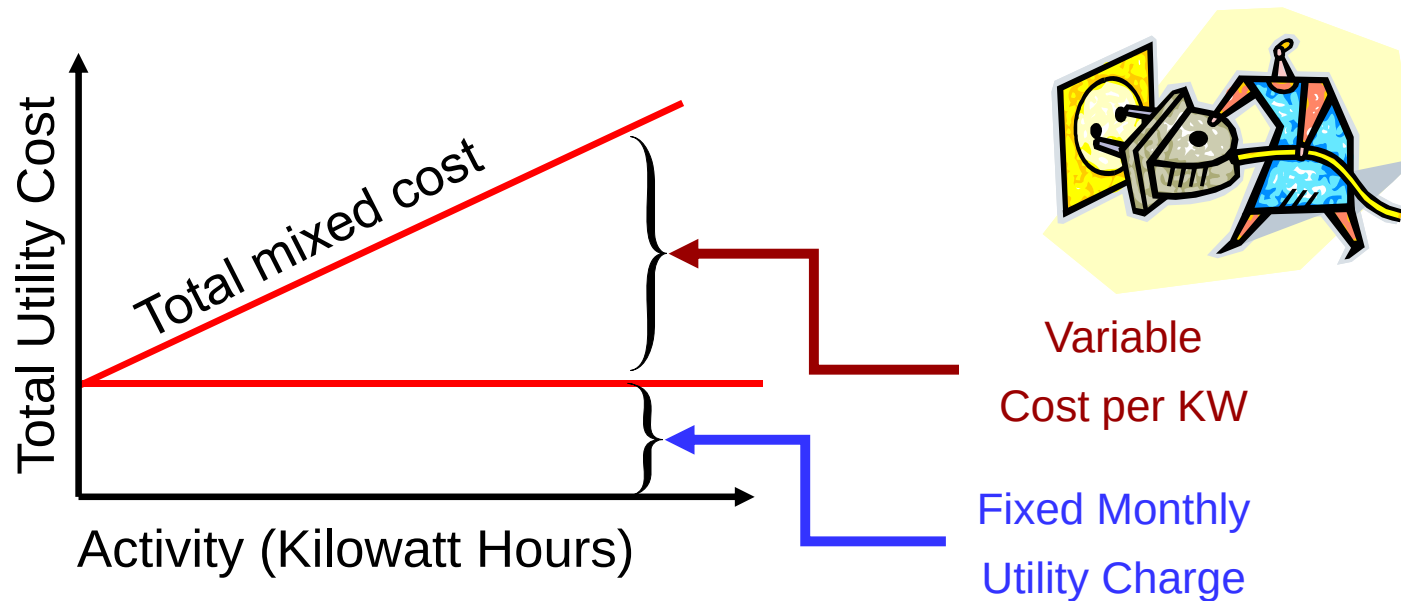
Total mixed costs increase as activity increases.



Per unit mixed costs decrease as activity increases.

Mixed costs

Mixed costs contain a fixed portion that is incurred even when facility is unused, and a variable portion that increases with usage. Utilities typically behave in this manner.



Cost Equation—Mixed Costs

Total mixed costs increase when volume increases, but **not** in direct proportion.

$$y = vx + f,$$

where

y = total mixed costs

v = variable cost per unit

x = number of variable units

f = fixed cost component

Key Characteristics of Mixed Costs

- Total mixed costs increase as volume increases because of the variable cost component
- Mixed costs *per unit* decrease as volume increases because of the fixed cost component
- Total mixed costs graphs slope upward but do *not* begin at the origin—they intersect the y-axis at the level of fixed costs
- Total mixed costs can be expressed as a *combination* of the variable and fixed cost equations:

Total mixed costs = variable cost component + fixed cost component

$$y = vx + f$$

where,

y = total mixed costs

v = variable cost per unit of activity (slope)

x = volume of activity

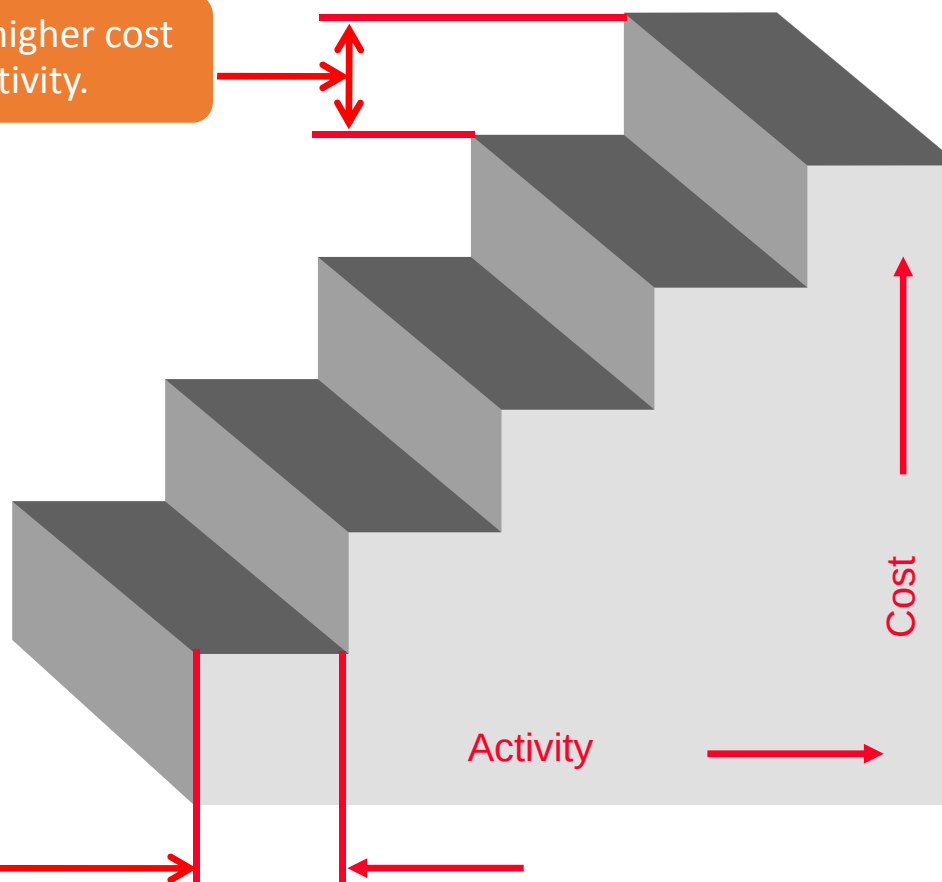
f = fixed cost over a given period of time (vertical intercept)

Example: Child care centers where ratio of teachers to students is regulated.

Step costs

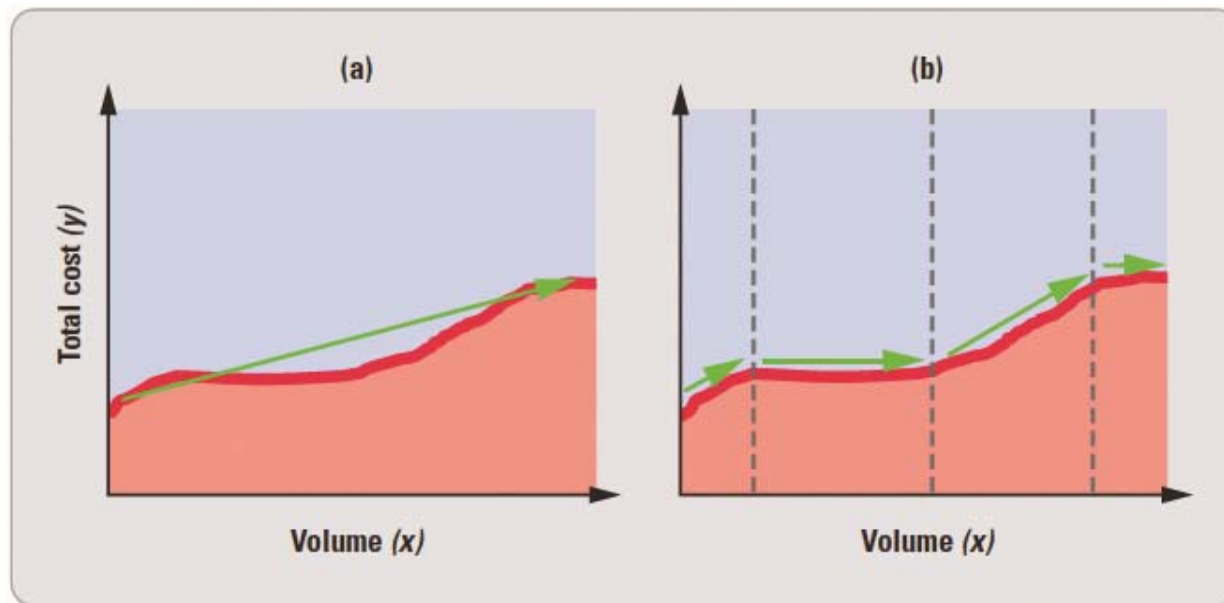
Total cost increases to a new higher cost for the next higher range of activity.

Total cost remains constant within a range of activity.



Other Cost Behaviors—Curvilinear Costs

- **Curvilinear costs:** not linear, do not fit into any neat pattern



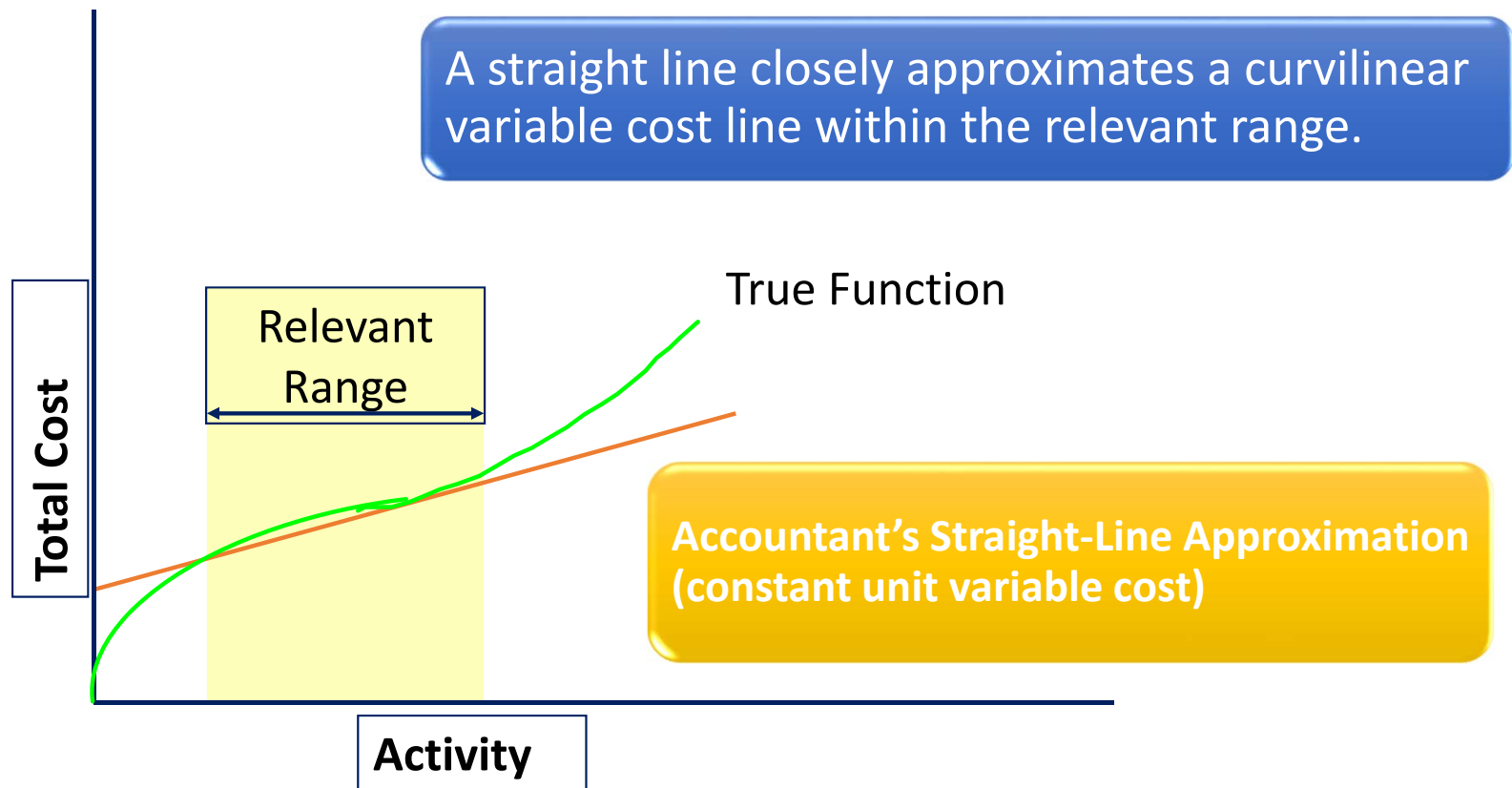
Constraints

Relatively short time horizon	Generally less than 1 year.
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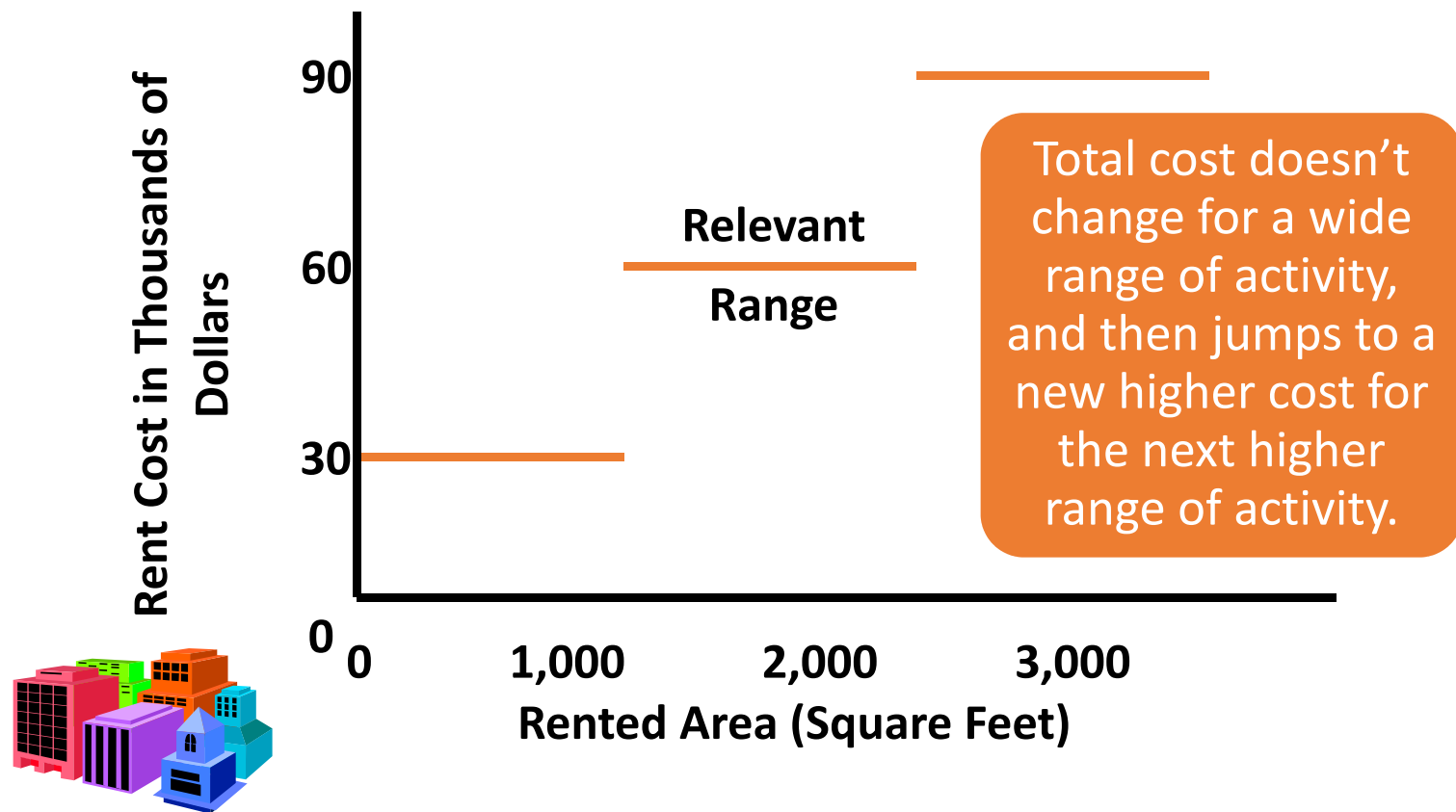
Relevant range	A limited range of activity over which we expect our assumptions about cost behavior to hold true.
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	E.g. We typically assume that the relationship between total cost and activity is LINEAR!
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The Linearity Assumption and the Relevant Range



Fixed Costs and Relevant Range



Relevant Range

The band of volume where the following remain constant:

- Total fixed costs
- Variable cost per unit